Supplementary materials

Clinical utility of circulating PTX3 as a prognostic biomarker in COVID-19: a systematic review and meta-analysis

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Yani Ke and Kaihan Wu are co-first authors of the article.

Jie Hu and Shan Liu are co-corresponding authors of the article.

Captions of supplementary materials

Supplementary material 1

PROSPERO with number CRD42022339907

Supplementary material 3

Search strategy

Supplementary material 3

(a) Sensitivity value of PTX3 for mortality. (b) Specificity value of PTX3 for mortality. (c) Diagnostic odds ratio (DOR) of PTX3 for mortality. (d) Positive likelihood ratio (PLR) of PTX3 for mortality. (e)Negative likelihood ratio (NLR) of PTX3 for mortality.

Supplementary material 4

PRISMA 2020 Checklist

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PROSPERO
International prospective register of systematic reviews

Circulating PTX3 levels as biomarkers for severity and mortality of COVID-19: a systematic review and meta-analysis

Citation

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Review question

Are circulating PTX3 levels correlated with severity and mortality of COVID-19?

Searches [1 change]

Sources: PubMed, EMBASE, Cochrane Library, ClinicalTrials.gov and grey literature

Language: English

Publication period: up to June, 2022

Search strategy: (“COVID-19” OR “COVID19” OR “SARS-CoV-2” OR “Sars-CoV-2 infection” OR “2019 nCoV” OR “2019-nCoV infection” OR “coronavirus” OR “coronavirus disease 2019” OR “nCoV pneumonia” OR “corona-virus”) AND (“ptx3” OR “pentraxin 3” OR “ptx-3” OR “pentraxin-3”)

Types of study to be included

Case-control studies or cohort studies or cross-sectional studies.

Condition or domain being studied

The outbreak of COVID-19 has become a hot topic in today's research. Although the pathological mechanism of intrapulmonary or extrapulmonary manifestations has been gradually clarified, there are still many uncertainties. SARS-CoV2 shows mild to moderate symptoms and can develop into severe cases, such as severe pneumonia to acute respiratory distress syndrome (ARDS), which is the main cause of death. Therefore, it is very important to find reasonable biomarkers to diagnose and detect the disease.

Pentraxin 3 (PTX3) is expressed in many immune cells and non-immune cells, produced mainly by vascular cell walls, which plays an irreplaceable role in tissue injury, wound, infection and wound healing. Furthermore, PTX3 levels are increasingly considered as a significant factor of autoimmune diseases development, which are speculated to be a key indicator for clinical practice of COVID-19. We also found that many studies have explored the relationship between PTX3 and COVID-19. The results of these studies are not exactly the same, or even the opposite, so we intend to analyse these data to draw more accurate conclusions.

Participants/population

Inclusion criteria: conducted in adults (aged 18 years or older); patients must be diagnosed with COVID-19; articles that focused on the circulating levels of PTX3.

Intervention(s), exposure(s)

Individuals with / without COVID-19, different severity of COVID-19, circulating PTX3 levels.

Comparator(s)/control

Healthy people without COVID-19 or patients with different severity of COVID-19.

Main outcome(s)

The relationship between circulating PTX3 levels and patients with COVID-19; the relationship between circulating PTX3 levels and severity in patients with COVID-19.

Additional outcome(s)

The relationship between circulating PTX3 levels and mortality in patients with COVID-19

the relationship between circulating PTX3 levels and prognosis in patients with COVID-19.

Data extraction (selection and coding)

Two researchers will extract data independently. Any disputed part will be arbitrated by a third researcher. The extracted information will include the first author's last name, date of publication, the original country / region, Newcastle-Ottawa Scale (NOS) determination, the number of cases and controls, measured method, diagnostic method, circulating PTX3 levels in each group, prognosis or mortality of patients.

Risk of bias (quality) assessment

The quality evaluation will mainly adopt the Newcastle-Ottawa Scale (NOS) scale. Two members independently assessed the quality of the included articles. Likewise, if there is a dispute, the decision would be left to a third researcher. The NOS scale can evaluate the selectivity, comparability and exposure of each literature. The publication bias of our study will be measured by the outcomes of Egger's test and funnel plot.

Strategy for data synthesis

Review Manager 5.3 and Stata 16 software will be utilized for this analysis. We will determine the effect model according to the heterogeneity among studies. If there is no significant heterogeneity among various studies (I^2 ≤ 50%), the fixed-effect model would be chosen. When the heterogeneity among studies is significant (I^2 > 50%), the random-effect model would be preferred. Subgroup analysis and meta-regression will be good methods to explore the sources of high heterogeneity.

The publication bias of our study will be measured by the outcomes of Egger's test and funnel plot.

Analysis of subgroups or subsets

Subgroups classified by different severity of COVID-19 or by different ethnicities / countries.

Contact details for further information

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Organisational affiliation of the review

Zhejiang Chinese Medical University

Review team members and their organisational affiliations

Miss KE YANI. Zhejiang Chinese Medical University

Dr Kaihan Wu.

Type and method of review

Epidemiologic, Meta-analysis, Prognostic, Systematic review

Anticipated or actual start date

30 June 2022

Anticipated completion date

31 August 2022

Funding sources/sponsors

None

Conflicts of interest

Language

English

Country

China

Stage of review

Review Ongoing

Subject index terms status

Subject indexing assigned by CRD

Subject index terms

Adult; Biomarkers; COVID-19; Disease Progression; Humans; Mortality; Prognosis; SARS-CoV-2; Tumor Necrosis Factors

Date of registration in PROSPERO

22 June 2022

Date of first submission

15 June 2022

Stage of review at time of this submission

The review has not started

Stage	Started	Completed
Preliminary searches	No	No
Piloting of the study selection process	No	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

The record owner confirms that the information they have supplied for this submission is accurate and complete and they understand that deliberate provision of inaccurate information or omission of data may be construed as scientific misconduct.

The record owner confirms that they will update the status of the review when it is completed and will add publication details in due course.

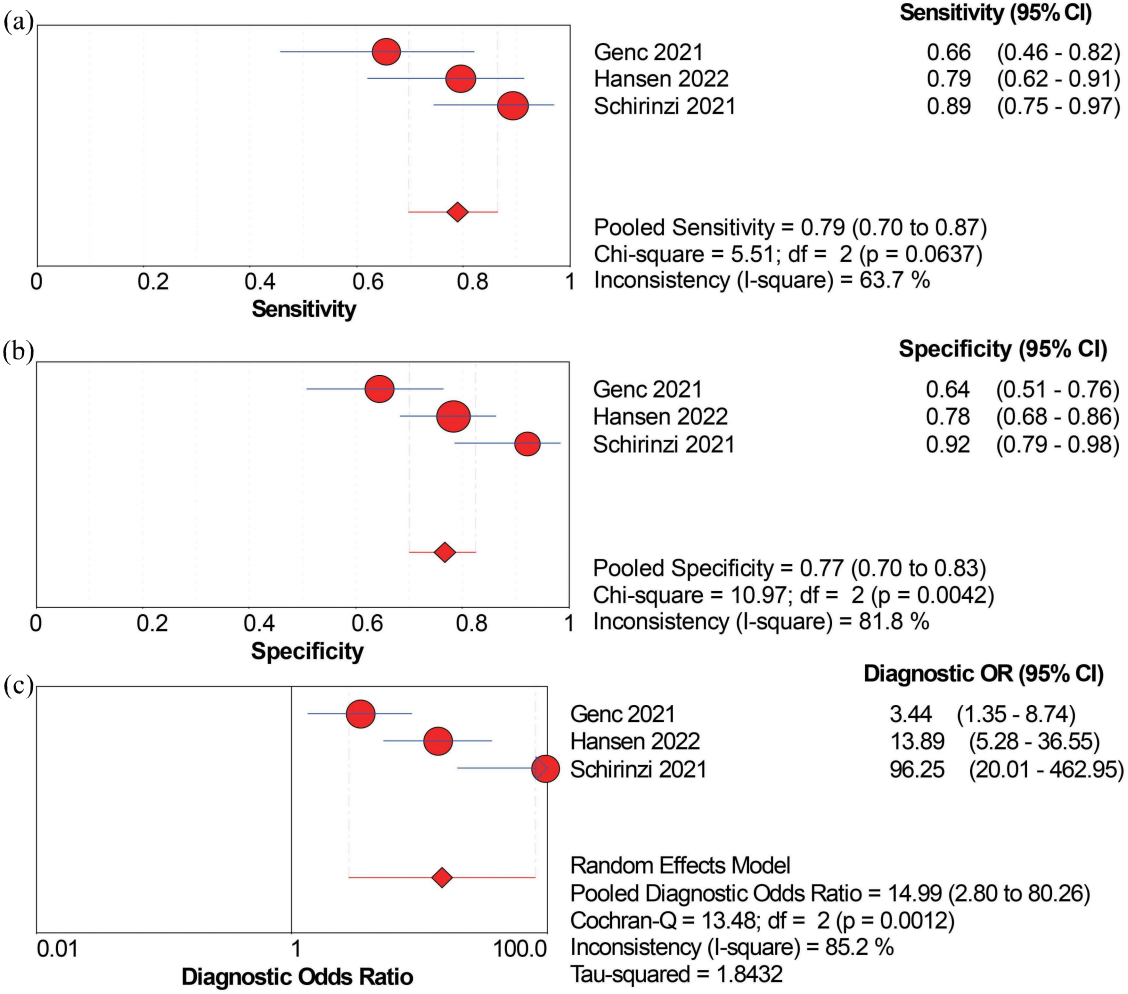
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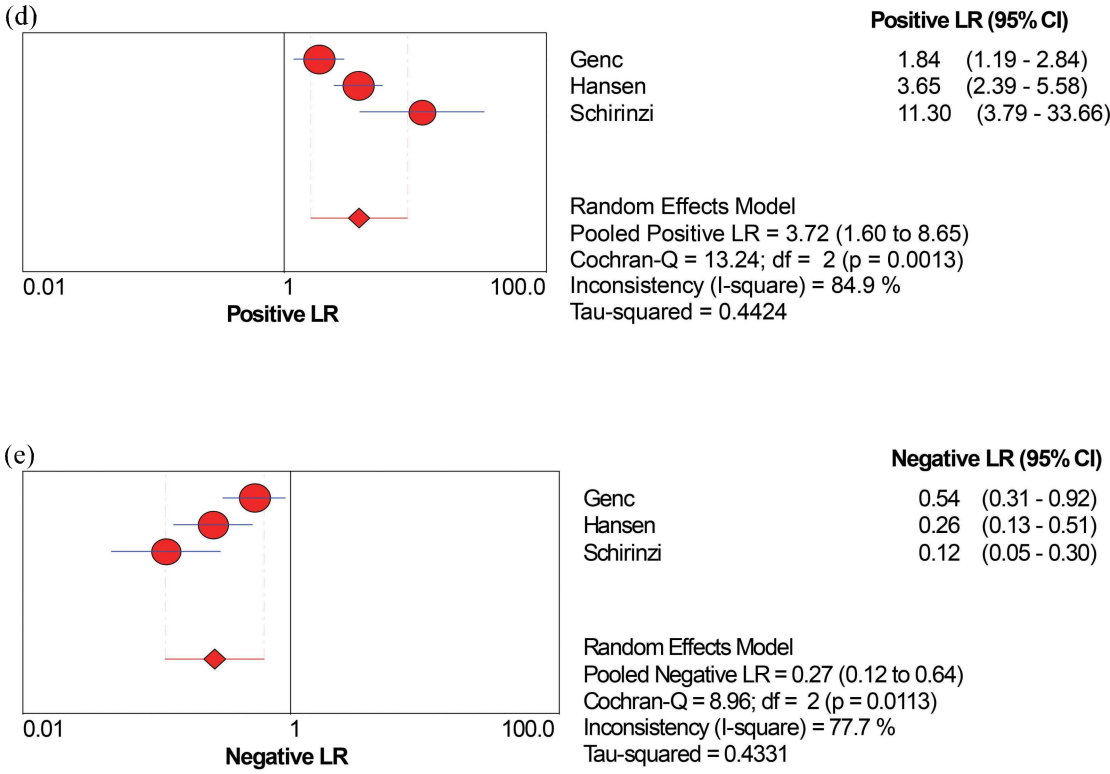
22 June 2022

Supplementary Material 2

Databases	Search terms	Number of records
Pubmed	Search：(('COVID-19')) OR ('COVID19') OR ('SARS-CoV-2') OR ('Sars-CoV-2 infection') OR ('2019 nCoV') OR ('2019-nCoV infection') OR ('coronavirus') OR ('coronavirus disease 2019') OR ('SARS-CoV-2') OR ('Novel coronavirus') OR ('nCoV') OR ('nCoV pneumonia') OR ('Emerging Coronavirus') OR ('new coronavirus') OR ('corona-virus') OR ('COVID-19'[MeSH Terms])) AND (('PTX3 protein')) OR ('pentraxin-3') OR ('pentraxin 3') OR ('ptx-3') OR ('ptx3') OR ('ptx3'[MeSH Terms]))	28
Cochrane Library	#1 MeSH descriptor:[pentraxin 3] explode all trees #2 'ptx3' #3 'ptx-3' #4 'pentraxin 3' #5 'pentraxin-3' #6 'PTX3 protein' #7 #1 or #2 or #3 or #4 or #5 or #6 #8 MeSH descriptor:[COVID-19] explode all trees #9 'COVID-19' #10 'COVID19' #11 'SARS-CoV-2' #12 'Sars-CoV-2 infection' #13 '2019 nCoV' #14 '2019-nCoV infection' #15'coronavirus' #16 'coronavirus disease 2019' #17 'nCoV pneumonia' #18 'corona-virus' #19 #10 or #11 or #12 or #13 or #14 or #15 or # 16 or #17 or #18 or #8 or #9 #20 #7 and #19	38
EMBASE	#1 'pentraxin 3'/exp #2 'ptx3' #3 'ptx-3' #4 'pentraxin 3' #5 'pentraxin-3' #6 'PTX3 protein' #7 #1 or #2 or #3 or #4 or #5 or #6 #8 'COVID-19'/exp #9 'COVID-19' #10 'COVID19' #11 'SARS-CoV-2' #12 'Sars-CoV-2 infection' #13 '2019 nCoV' #14 '2019-nCoV infection' #15 'coronavirus' #16 'coronavirus disease 2019' #17 'nCoV pneumonia' #18 'corona-virus' #19 #10 or # 11 or #12 or #13 or #14 or 15 or #16 or #17 or #18 or #9 or #8 #20 #19 and #7	58

Supplementary Material 3





Supplementary Material 4



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	7-8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	7-8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	7-8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	5-7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	7-8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	7-8
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	7-8
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	7-8
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	7-8
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	8
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	7-8

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	8-9
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	8-9
Study characteristics	17	Cite each included study and present its characteristics.	9-11
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	11
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	9-11,22-24
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	9-11
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	9-11
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	9-11
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	11
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	9-10
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	11-12
	23b	Discuss any limitations of the evidence included in the review.	12
	23c	Discuss any limitations of the review processes used.	12
	23d	Discuss implications of the results for practice, policy, and future research.	11-15
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	5
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	16
Competing interests	26	Declare any competing interests of review authors.	16
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	16

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71